Supplemental Digital Content 3 – Table listing the studies that were included as part of the systematic review

1	Zhang Y, Fatemi P, Medress Z, et al. A predictive-modeling based screening tool for prolonged opioid use after surgical management of low back and lower extremity pain. *Spine J* . 2020;20(8):1184-1195. doi:10.1016/j.spinee.2020.05.098
2	Welk B, McClure JA, Clarke C, Vogt K, Campbell J. An Opioid Prescription for Men Undergoing Minor Urologic Surgery Is Associated with an Increased Risk of New Persistent Opioid Use. *Eur Urol* . 2020;77(1):68-75. doi:10.1016/j.eururo.2019.08.031
3	Reyes AA, Canseco JA, Mangan JJ, et al. Risk Factors for Prolonged Opioid Use and Effects of Opioid Tolerance on Clinical Outcomes after Anterior Cervical Discectomy and Fusion Surgery. *Spine (Phila Pa 1976)* . 2020;45(14):968-975. doi:10.1097/BRS.0000000000003511
4	Reid DBC, Shah KN, Shapiro BH, et al. Opioid-Limiting Legislation Associated With Reduced Postoperative Prescribing After Surgery for Traumatic Orthopaedic Injuries. *J Orthop Trauma* . 2020;34(4):e114-e120. doi:10.1097/BOT.0000000000001673
5	Labaran L, Aryee JNA, Bell J, et al. Opioids and spinal cord stimulators: Pre-and postoperative opioid use patterns and predictors of prolonged postoperative opioid use. *Neurospine* . 2020;17(1):246-253. doi:10.14245/ns.1938308.154
6	Anciano Granadillo V, Cancienne JM, Gwathmey FW, Werner BC. Perioperative Opioid Analgesics and Hip Arthroscopy: Trends, Risk Factors for Prolonged Use, and Complications. *Arthrosc - J Arthrosc Relat Surg* . 2018;34(8):2359-2367. doi:10.1016/j.arthro.2018.03.016
7	Young JC, Wu JM, Willis-Gray M, Pate V, Jonsson Funk M. Persistent Opioid Use after Hysterectomy in the United States, 2005-2015. *Obstet Gynecol* . 2020;135(1):123-132. doi:10.1097/AOG.0000000000003612
8	Bennett KG, Harbaugh CM, Hu HM, et al. Persistent opioid use among children, adolescents, and young adults after common cleft operations. *J Craniofac Surg* . 2018;29(7):1697-1701. doi:10.1097/SCS.0000000000004762
9	Harbaugh CM, Lee JS, Hu HM, et al. Persistent opioid use among pediatric patients after surgery. *Pediatrics* . 2018;141(1):1-11. doi:10.1542/peds.2017-2439
10	Nelson DB, Niu J, Mitchell KG, et al. Persistent Opioid Use Among the Elderly After Lung Resection: A SEER-Medicare Study. *Ann Thorac Surg* . 2020;109(1):194-202. doi:10.1016/j.athoracsur.2019.06.095
11	Bennett KG, Kelley BP, Vick AD, et al. Persistent opioid use and high-risk prescribing in body contouring patients. *Plast Reconstr Surg* . 2019;143(1):87-96. doi:10.1097/PRS.0000000000005084
12	Anderson AB, Grazal CF, Balazs GC, Potter BK, Dickens JF, Forsberg JA. Can Predictive Modeling Tools Identify Patients at High Risk of Prolonged Opioid Use After ACL Reconstruction? *Clin Orthop Relat Res* . 2020;478(7):00-1618. doi:10.1097/corr.0000000000001251
13	Nelson DB, Cata JP, Niu J, et al. Persistent opioid use is associated with worse survival after lobectomy for stage i non-small cell lung cancer. *Pain* . 2019;160(10):2365-2373. doi:10.1097/j.pain.0000000000001630
14	Schoenfeld AJ, Nwosu K, Jiang W, et al. Risk Factors for Prolonged Opioid Use Following Spine Surgery, and the Association with Surgical Intensity, among Opioid-Naive Patients. *J Bone Jt Surg - Am Vol* . 2017;99(15):1247-1252. doi:10.2106/JBJS.16.01075
15	Saraswathula A, Chen MM, Mudumbai SC, Whittemore AS, Divi V. Persistent Postoperative Opioid Use in Older Head and Neck Cancer Patients. *Otolaryngol - Head Neck Surg (United States)* . 2019;160(3):380-387. doi:10.1177/0194599818778276
16	Brock JL, Jain N, Phillips FM, Malik AT, Khan SN. Postoperative opioid cessation rates based on preoperative opioid use: Analysis of common orthopaedic procedures. *Bone Jt J* . 2019;101-B(12):1570-1577. doi:10.1302/0301-620X.101B12.BJJ-2019-0080.R2
17	Rogero R, Fuchs D, Nicholson K, et al. Postoperative Opioid Consumption in Opioid-Naive Patients Undergoing Hallux Valgus Correction. *Foot Ankle Int* . 2019;40(11):1267-1272. doi:10.1177/1071100719862606
18	Young JC, Dasgupta N, Chidgey BA, Jonsson Funk M. Postsurgical Opioid Prescriptions and Risk of Long-term Use: An Observational Cohort Study Across the United States. *Ann Surg* . 2019;(Online ahead of print). doi:10.1097/SLA.0000000000003549
19	Karhade A V., Cha TD, Fogel HA, et al. Predicting prolonged opioid prescriptions in opioid-naive lumbar spine surgery patients. *Spine J* . 2020;20(6):888-895. doi:10.1016/j.spinee.2019.12.019
20	Hockley A, Ge D, Vasquez-Montes D, et al. Predictors of long-term opioid dependence in transforaminal lumbar interbody fusion with a focus on pre-operative opioid usage. *Eur Spine J* . 2020;29(6):1311-1317. doi:10.1007/s00586-020-06345-3
21	Connolly J, Javed Z, Raji MA, Chan W, Kuo YF, Baillargeon J. Predictors of Long-term Opioid Use Following Lumbar Fusion Surgery. *Spine (Phila Pa 1976)* . 2017;42(18):1405-1411. doi:10.1097/BRS.0000000000002133
22	Clement KC, Canner JK, Lawton JS, Whitman GJR, Grant MC, Sussman MS. Predictors of new persistent opioid use after coronary artery bypass grafting. *J Thorac Cardiovasc Surg* . 2020;160(4):954-963.e4. doi:10.1016/j.jtcvs.2019.09.137
23	Fields AC, Cavallaro PM, Correll DJ, et al. Predictors of Prolonged Opioid Use Following Colectomy. *Dis Colon Rectum* . 2019;62(9):1117-1123. doi:10.1097/DCR.0000000000001429
24	Wright AK, Sikora M, Leveque JC. Characterizing the Risk of Long-Term Opioid Utilization in Patients Undergoing Lumbar Spine Surgery. *Spine (Phila Pa 1976)* . 2020;45(1):E54-E60. doi:10.1097/BRS.0000000000003199
25	Yang S, Werner BC. Risk Factors for Prolonged Postoperative Opioid Use after Spinal Fusion for Adolescent Idiopathic Scoliosis. *J Pediatr Orthop* . 2019;39(10):500-504. doi:10.1097/BPO.0000000000001139
26	Jain N, Phillips FM, Weaver T, Khan SN. Preoperative chronic opioid therapy: A risk factor for complications, readmission, continued opioid use and increased costs after one- and two-level posterior lumbar fusion. *Spine (Phila Pa 1976)* . 2018;43(19):1331-1338. doi:10.1097/BRS.0000000000002609
27	Kim KY, Anoushiravani AA, Chen KK, Roof M, Long WJ, Schwarzkopf R. Preoperative Chronic Opioid Users in Total Knee Arthroplasty—Which Patients Persistently Abuse Opiates Following Surgery? *J Arthroplasty* . 2018;33(1):107-112. doi:10.1016/j.arth.2017.07.041
28	Hozack BA, Rivlin M, Lutsky KF, et al. Preoperative exposure to benzodiazepines or sedative/hypnotics increases the risk of greater filled opioid prescriptions after surgery. *Clin Orthop Relat Res* . 2019;477(6):1482-1488. doi:10.1097/CORR.0000000000000696
29	Rojas EO, Khazi ZM, Gulbrandsen TR, et al. Preoperative Opioid Prescription Filling Is a Risk Factor for Prolonged Opioid Use After Elbow Arthroscopy. *Arthrosc - J Arthrosc Relat Surg* . 2020;36(8):2106-2113. doi:10.1016/j.arthro.2020.04.053
30	Khazi ZM, Baron J, Shamrock A, et al. Preoperative Opioid Usage, Male Sex, and Preexisting Knee Osteoarthritis Impacts Opioid Refills After Isolated Arthroscopic Meniscectomy: A Population-Based Study. *Arthrosc - J Arthrosc Relat Surg* . 2020;36(9):2478-2485. doi:10.1016/j.arthro.2020.04.039
31	Khazi ZM, Shamrock AG, Hajewski C, et al. Preoperative opioid use is associated with inferior outcomes after patellofemoral stabilization surgery. *Knee Surgery, Sport Traumatol Arthrosc* . 2020;28(2):599-605. doi:10.1007/s00167-019-05738-2
32	Bell JE, Sequeira SB, Chen DQ, Haug EC, Werner BC, Browne JA. Preoperative Pain Management: Is Tramadol a Safe Alternative to Traditional Opioids Before Total Hip Arthroplasty? *J Arthroplasty* . 2020;35(10):2886-2891.e1. doi:10.1016/j.arth.2020.04.093
33	Prentice HA, Inacio MCS, Singh A, Namba RS, Paxton EW. Preoperative Risk Factors for Opioid Utilization after Total Hip Arthroplasty. *J Bone Jt Surg - Am Vol* . 2019;101(18):1670-1678. doi:10.2106/JBJS.18.01005
34	Basilico M, Bhashyam AR, Harris MB, Heng M. Prescription Opioid Type and the Likelihood of Prolonged Opioid Use After Orthopaedic Surgery. *J Am Acad Orthop Surg* . 2019;27(9):e423-e429. doi:10.5435/JAAOS-D-17-00663
35	Marcusa DP, Mann RA, Cron DC, et al. Prescription Opioid Use among Opioid-Naive Women Undergoing Immediate Breast Reconstruction. *Plast Reconstr Surg* . 2017;140(6):1081-1090. doi:10.1097/PRS.0000000000003832
36	Hirji SA, Landino S, Cote C, et al. Chronic opioid use after coronary bypass surgery. *J Card Surg* . 2019;34(2):67-73. doi:10.1111/jocs.13981
37	Hadlandsmyth K, Vander Weg MW, McCoy KD, Mosher HJ, Vaughan-Sarrazin MS, Lund BC. Risk for Prolonged Opioid Use Following Total Knee Arthroplasty in Veterans. *J Arthroplasty* . 2018;33(1):119-123. doi:10.1016/j.arth.2017.08.022
38	Zaveri S, Nobel TB, Khetan P, Divino CM. Risk of Chronic Opioid Use in Opioid-Naïve and Non-Naïve Patients after Ambulatory Surgery. *J Gastrointest Surg* . 2020;24(3):688-694. doi:10.1007/s11605-019-04265-2
39	Gil JA, Gunaseelan V, DeFroda SF, Brummett CM, Bedi A, Waljee JF. Risk of Prolonged Opioid Use Among Opioid-Naïve Patients After Common Shoulder Arthroscopy Procedures. *Am J Sports Med* . 2019;47(5):1043-1050. doi:10.1177/0363546518819780
40	Johnson SP, Chung KC, Zhong L, et al. Risk of Prolonged Opioid Use Among Opioid-Naïve Patients Following Common Hand Surgery Procedures. *J Hand Surg Am* . 2016;41(10):947-957.e3. doi:10.1016/j.jhsa.2016.07.113
41	Shkolyar E, Shih IF, Li Y, Wong JA, Liao JC. Robot-Assisted Radical Prostatectomy Associated with Decreased Persistent Postoperative Opioid Use. *J Endourol* . 2020;34(4):475-481. doi:10.1089/end.2019.0788
42	Barham DW, McMann LP, Musser JE, et al. Routine prescription of opioids for post-vasectomy pain control associated with persistent use. *J Urol* . 2019;202(4):806-810. doi:10.1097/JU.0000000000000304
43	Anderson JT, Haas AR, Percy R, Woods ST, Ahn UM, Ahn NU. Chronic opioid therapy after lumbar fusion surgery for degenerative disc disease in a workers' compensation setting. *Spine (Phila Pa 1976)* . 2015;40(22):1775-1784. doi:10.1097/BRS.0000000000001054
44	Adogwa O, Davison MA, Vuong V, et al. Sex Differences in Opioid Use in Patients With Symptomatic Lumbar Stenosis or Spondylolisthesis Undergoing Lumbar Decompression and Fusion. *Spine (Phila Pa 1976)* . 2019;44(13):E800-E807. doi:10.1097/BRS.0000000000002965
45	Yang DS, Li NY, Mariorenzi MC, Kleinhenz DT, Cohen EM, Daniels AH. Surgical Treatment of Patients With Dual Hip and Spinal Degenerative Disease: Effect of Surgical Sequence of Spinal Fusion and Total Hip Arthroplasty on Postoperative Complications. *Spine (Phila Pa 1976)* . 2020;45(10):E587-E593. doi:10.1097/BRS.0000000000003351
46	Katzman C, Harker EC, Ahmed R, et al. The Association Between Preoperative Opioid Exposure and Prolonged Postoperative Use. *Ann Surg* . 2020;(Online ahead of print). doi:10.1097/sla.0000000000003723
47	Steen T, Lirk PB, Sigurdsson MI. The demographics of persistent opioid consumption following limb amputation. *Acta Anaesthesiol Scand* . 2020;64(3):361-367. doi:10.1111/aas.13497
48	Harris AB, Marrache M, Jami M, et al. Chronic opioid use following anterior cervical discectomy and fusion surgery for degenerative cervical pathology. *Spine J* . 2020;20(1):78-86. doi:10.1016/j.spinee.2019.09.011
49	Grace TR, Khanna K, Choo KJ, et al. The influence of preoperative opioid use on inpatient opioid requirements and discharge prescriptions after primary shoulder arthroplasty. *J Shoulder Elb Surg* . 2018;27(9):1572-1579. doi:10.1016/j.jse.2018.02.067
50	Klueh MP, Hu HM, Howard RA, et al. Transitions of Care for Postoperative Opioid Prescribing in Previously Opioid-Naïve Patients in the USA: a Retrospective Review. *J Gen Intern Med* . Published online 2018:1-7. doi:10.1007/s11606-018-4463-1
51	Goesling J, Moser SE, Zaidi B, et al. Trends and predictors of opioid use after total knee and total hip arthroplasty. *Pain* . 2016;157(6):1259-1265. doi:10.1097/j.pain.0000000000000516
52	Politzer CS, Kildow BJ, Goltz DE, Green CL, Bolognesi MP, Seyler TM. Trends in Opioid Utilization Before and After Total Knee Arthroplasty. *J Arthroplasty* . 2018;33(7):S147-S153.e1. doi:10.1016/j.arth.2017.10.060
53	Tan TL, Rondon AJ, Wilt Z, et al. Understanding Opioid Use After Total Hip Arthroplasty: A Comprehensive Analysis of a Mandatory Prescription Drug Monitoring Program. *J Am Acad Orthop Surg* . 2020;28(20):e917-e922. doi:10.5435/JAAOS-D-19-00676
54	Cron DC, Tincopa MA, Lee JS, et al. Prevalence and Patterns of Opioid Use Before and After Liver Transplantation. *Transplantation* . 2021;105(1):100-107. doi:10.1097/tp.0000000000003155
55	Kuo JH, Huang Y, Kluger MD, et al. Use and Misuse of Opioids After Endocrine Surgery Operations. *Ann Surg* . 2020;(Online ahead of print). doi:10.1097/sla.0000000000003777
56	Wright JD, Huang Y, Melamed A, et al. Use and Misuse of Opioids After Gynecologic Surgical Procedures. *Obstet Gynecol* . 2019;134(2):250-260. doi:10.1097/AOG.0000000000003358
57	Deyo RA, Hallvik SE, Hildebran C, et al. Use of prescription opioids before and after an operation for chronic pain (lumbar fusion surgery). *Pain* . 2018;159(6):1147-1154. doi:10.1097/j.pain.0000000000001202
58	Karhade A V., Chaudhary MA, Bono CM, Kang JD, Schwab JH, Schoenfeld AJ. Validating the Stopping Opioids after Surgery (SOS) score for sustained postoperative prescription opioid use in spine surgical patients. *Spine J* . 2019;19(10):1666-1671. doi:10.1016/j.spinee.2019.05.001
59	Stafford C, Francone T, Roberts PL, Ricciardi R. What factors are associated with increased risk for prolonged postoperative opioid usage after colorectal surgery? *Surg Endosc* . 2018;32(8):3557-3561. doi:10.1007/s00464-018-6078-3
60	Pang J, Tringale KR, Tapia VJ, et al. Chronic opioid use following surgery for oral cavity cancer. *JAMA Otolaryngol - Head Neck Surg* . 2017;143(12):1187-1194. doi:10.1001/jamaoto.2017.0582
61	Wen X, Kogut S, Aroke H, Taylor L, Matteson KA. Chronic opioid use in women following hysterectomy: Patterns and predictors. *Pharmacoepidemiol Drug Saf* . 2020;29(4):493-503. doi:10.1002/pds.4972
62	Thiels CA, Habermann EB, Hooten WM, Jeffery MM. Chronic use of tramadol after acute pain episode: Cohort study. *BMJ* . 2019;365:18-20. doi:10.1136/bmj.l1849
63	Swenson CW, Kamdar NS, Seiler K, Morgan DM, Lin P, As-Sanie S. Definition development and prevalence of new persistent opioid use following hysterectomy. *Am J Obstet Gynecol* . 2018;219(5):486.e1-486.e7. doi:10.1016/j.ajog.2018.06.010
64	Chaudhary MA, Bhulani N, De Jager EC, et al. Development and Validation of a Bedside Risk Assessment for Sustained Prescription Opioid Use after Surgery. *JAMA Netw Open* . 2019;2(7):1-10. doi:10.1001/jamanetworkopen.2019.6673
65	Stark N, Kerr S, Stevens J. Prevalence and predictors of persistent post-surgical opioid use: a prospective observational cohort study. *Anaesth Intensive Care* . 2017;45(6):700-706. doi:10.1177/0310057x1704500609
66	Katakam A, Karhade A V., Schwab JH, Chen AF, Bedair HS. Development and validation of machine learning algorithms for postoperative opioid prescriptions after TKA. *J Orthop* . 2020;22(March):95-99. doi:10.1016/j.jor.2020.03.052
67	Nicholson AD, Kassam HF, Steele JL, Passarelli NR, Blaine TA, Kovacevic D. Development of a clinical risk calculator for prolonged opioid use after shoulder surgery. *J Shoulder Elb Surg* . 2019;28(11):2225-2231. doi:10.1016/j.jse.2019.03.033
68	Harris AB, Marrache M, Puvanesarajah V, et al. Are preoperative depression and anxiety associated with patient-reported outcomes, health care payments, and opioid use after anterior discectomy and fusion? *Spine J* . 2020;20(8):1167-1175. doi:10.1016/j.spinee.2020.03.004
69	Karhade A V., Ogink PT, Thio QCBS, et al. Development of machine learning algorithms for prediction of prolonged opioid prescription after surgery for lumbar disc herniation. *Spine J* . 2019;19(11):1764-1771. doi:10.1016/j.spinee.2019.06.002
70	Karhade A V., Schwab JH, Bedair HS. Development of Machine Learning Algorithms for Prediction of Sustained Postoperative Opioid Prescriptions After Total Hip Arthroplasty. *J Arthroplasty* . 2019;34(10):2272-2277.e1. doi:10.1016/j.arth.2019.06.013
71	Ly DP. Differences Within Practices in Opioid-Prescribing Patterns of Orthopedic Surgeons and in Subsequent Rates of Chronic Opioid Use, 2012–2014. *J Gen Intern Med* . 2019;34(4):529-531. doi:10.1007/s11606-018-4745-7
72	Agarwalla A, Liu JN, Gowd AK, Amin NH, Werner BC. Differential Use of Narcotics in Total Hip Arthroplasty: A Comparative Matched Analysis Between Osteoarthritis and Femoral Neck Fracture. *J Arthroplasty* . 2020;35(2):471-476. doi:10.1016/j.arth.2019.09.004
73	Neuman MD, Hennessy S, Small DS, et al. Drug Enforcement Agency 2014 Hydrocodone Rescheduling Rule and Opioid Dispensing after Surgery. *Anesthesiology* . 2020;(5):1151-1164. doi:10.1097/ALN.0000000000003188
74	Reid DBC, Shah KN, Ruddell JH, et al. Effect of narcotic prescription limiting legislation on opioid utilization following lumbar spine surgery. *Spine J* . 2019;19(4):717-725. doi:10.1016/j.spinee.2018.09.007
75	Smith ME, Lee JS, Bonham A, et al. Effect of new persistent opioid use on physiologic and psychologic outcomes following bariatric surgery. *Surg Endosc* . 2019;33(8):2649-2656. doi:10.1007/s00464-018-6542-0
76	Wu L, Li M, Zeng Y, et al. Prevalence and risk factors for prolonged opioid use after total joint arthroplasty: a systematic review, meta-analysis, and meta-regression. *Arch Orthop Trauma Surg* . 2020;(Online ahead of print). doi:10.1007/s00402-020-03486-4
77	Leroux TS, Saltzman BM, Sumner SA, et al. Elective Shoulder Surgery in the Opioid Naïve: Rates of and Risk Factors for Long-term Postoperative Opioid Use. *Am J Sports Med* . 2019;47(5):1051-1056. doi:10.1177/0363546519837516
78	Wang MC, Lozen AM, Laud PW, Nattinger AB, Krebs EE. Factors associated with chronic opioid use after cervical spine surgery for degenerative conditions. *J Neurosurg Spine* . 2020;32(1):1-8. doi:10.3171/2019.7.SPINE19563
79	Brescia AA, Harrington CA, Mazurek AA, et al. Factors Associated With New Persistent Opioid Usage After Lung Resection. *Ann Thorac Surg* . 2019;107(2):363-368. doi:10.1016/j.athoracsur.2018.08.057
80	Harbaugh CM, Lee JS, Chua KP, et al. Association between Long-term Opioid Use in Family Members and Persistent Opioid Use after Surgery among Adolescents and Young Adults. *JAMA Surg* . 2019;154(4):1-9. doi:10.1001/jamasurg.2018.5838
81	Sharma M, Ugiliweneza B, Aljuboori Z, Nuño MA, Drazin D, Boakye M. Factors predicting opioid dependence in patients undergoing surgery for degenerative spondylolisthesis: analysis from the MarketScan databases. *J Neurosurg Spine* . 2018;29(3):271-278. doi:10.3171/2018.1.spine171258
82	Dy CJ, Peacock K, Olsen MA, Ray WZ, Brogan DM. Frequency and Risk Factors for Prolonged Opioid Prescriptions After Surgery for Brachial Plexus Injury. *J Hand Surg Am* . 2019;44(8):662-668.e1. doi:10.1016/j.jhsa.2019.04.001
83	Brummett CM, England C, Evans-Shields J, et al. Health care burden associated with outpatient opioid use following inpatient or outpatient surgery. *J Manag Care Spec Pharm* . 2019;25(9):973-983. doi:10.18553/jmcp.2019.19055
84	Lee JS, Vu J V., Edelman AL, et al. Health Care Spending and New Persistent Opioid Use After Surgery. *Ann Surg* . 2020;272(1):99-104. doi:10.1097/SLA.0000000000003399
85	Sharma M, Ugiliweneza B, Aljuboori Z, Boakye M. Health care utilization and overall costs based on opioid dependence in patients undergoing surgery for degenerative spondylolisthesis. *Neurosurg Focus* . 2018;44(5):1-10. doi:10.3171/2018.2.FOCUS17764
86	Delaney LD, Gunaseelan V, Rieck H, Dupree JM, Hallstrom BR, Waljee JF. High-Risk Prescribing Increases Rates of New Persistent Opioid Use in Total Hip Arthroplasty Patients. *J Arthroplasty* . 2020;35(9):2472-2479.e2. doi:10.1016/j.arth.2020.04.019
87	Rozet I, Nishio I, Robbertze R, Rotter D, Chansky H, Hernandez A V. Prolonged opioid use after knee arthroscopy in military veterans. *Anesth Analg* . 2014;119(2):454-459. doi:10.1213/ANE.0000000000000292
88	Chatha K, Borroto W, Goss L, Ghisa C, Gilot G, Sabesan VJ. How orthopedic surgeons can impact opioid use and dependence in shoulder arthroplasty. *JSES Int* . 2020;4(1):105-108. doi:10.1016/j.jses.2019.10.113
89	Woeste MR, Bhutiani N, Geller AE, Eldridge-Hindy H, McMasters KM, Ajkay N. Identifying Factors Predicting Prolonged Opioid Use After Mastectomy. *Ann Surg Oncol* . 2020;27(4):993-1001. doi:10.1245/s10434-019-08171-4
90	Chen Q, Hsia HL, Overman R, et al. Impact of an Opioid Safety Initiative on Patients Undergoing Total Knee Arthroplasty: A Time Series Analysis. *Anesthesiology* . 2019;131(2):369-380. doi:10.1097/ALN.0000000000002771
91	Kim Y, Cortez AR, Wima K, et al. Impact of Preoperative Opioid Use After Emergency General Surgery. *J Gastrointest Surg* . 2018;22(6):1098-1103. doi:10.1007/s11605-017-3665-x
92	Bicket MC, Murimi IB, Mansour O, Wu CL, Alexander GC. Association of new opioid continuation with surgical specialty and type in the United States. *Am J Surg* . 2019;218(5):818-827. doi:10.1016/j.amjsurg.2019.04.010
93	Brescia AA, Waljee JF, Hu HM, et al. Impact of Prescribing on New Persistent Opioid Use After Cardiothoracic Surgery. *Ann Thorac Surg* . 2019;108(4):1107-1113. doi:10.1016/j.athoracsur.2019.06.019
94	Eley N, Sikora M, Wright AK, Leveque JC. Implementation of an Opioid Reduction Protocol for Simple Outpatient Neurosurgical Procedures: A Single-Center Experience. *Spine (Phila Pa 1976)* . 2020;45(6):397-404. doi:10.1097/BRS.0000000000003272
95	Jafari A, Shen SA, Bracken DJ, Pang J, DeConde AS. Incidence and predictive factors for additional opioid prescription after endoscopic sinus surgery. *Int Forum Allergy Rhinol* . 2018;8(8):883-889. doi:10.1002/alr.22150
96	Dunn LK, Yerra S, Fang S, et al. Incidence and risk factors for chronic postoperative opioid use after major spine surgery: A cross-sectional study with longitudinal outcome. *Anesth Analg* . 2018;127(1):247-254. doi:10.1213/ANE.0000000000003338
97	Ward A, De Souza E, Miller D, et al. Incidence of and Factors Associated with Prolonged and Persistent Postoperative Opioid Use in Children 0-18 Years of Age. *Anesth Analg* . 2020;131(4):1237-1248. doi:10.1213/ANE.0000000000004823
98	Shen C, Thornton JD, Gu D, et al. Prolonged Opioid Use After Surgery for Early-Stage Breast Cancer. *Oncologist* . 2020;25(10):1-9. doi:10.1634/theoncologist.2019-0868
99	Sun EC, Darnall BD, Baker LC, MacKey S. Incidence of and risk factors for chronic opioid use among opioid-naive patients in the postoperative period. *JAMA Intern Med* . 2016;176(9):1286-1293. doi:10.1001/jamainternmed.2016.3298
100	Li NY, Patel SA, Durand WM, Ready L V., Owens BD, Daniels AH. Increased Risk of Chronic Opioid Use and Revision After Anterior Cervical Diskectomy and Fusion in Patients with Prior Shoulder Arthroscopy. *World Neurosurg* . 2020;135:e202-e208. doi:10.1016/j.wneu.2019.11.116
101	Wilson LA, Liu J, Fiasconaro M, Poeran J, Nwachukwu BU, Memtsoudis SG. Increased Use of Intra-Articular Steroid Injection to Treat Osteoarthritis is Associated With Chronic Opioid Dependence After Later Total Knee Arthroplasty But Not Total Hip Arthroplasty. *J Arthroplasty* . 2020;35(8):1979-1982. doi:10.1016/j.arth.2020.04.012
102	Sun EC, Bateman BT, Memtsoudis SG, Neuman MD, Mariano ER, Baker LC. Lack of Association between the Use of Nerve Blockade and the Risk of Postoperative Chronic Opioid Use among Patients Undergoing Total Knee Arthroplasty: Evidence from the Marketscan Database. *Anesth Analg* . 2017;125(3):999-1007. doi:10.1213/ANE.0000000000001943
103	Alam A, Gomes T, Zheng H, Mamdani MM, Juurlink DN, Bell CM. Long-term analgesic use after low-risk surgery: A retrospective cohort study. *Arch Intern Med* . 2012;172(5):425-430. doi:10.1001/archinternmed.2011.1827
104	Lanzillotta-Rangeley J, Clark A, Christianson A, Kalarchian MA. Association of prescription opioid exposure and patient factors with prolonged postoperative opioid use in opioid-naive patients. *AANA J* . 2020;88(1):18-26.
105	Maciejewski ML, Smith VA, Berkowitz TSZ, et al. Long-term opioid use after bariatric surgery. *Surg Obes Relat Dis* . 2020;16(8):1100-1110. doi:10.1016/j.soard.2020.04.037
106	Bartels K, Fernandez-Bustamante A, McWilliams SK, Hopfer CJ, Mikulich-Gilbertson SK. Long-term opioid use after inpatient surgery – A retrospective cohort study. *Drug Alcohol Depend* . 2018;187:61-65. doi:10.1016/j.drugalcdep.2018.02.013
107	Karhade A V., Ogink PT, Thio QCBS, et al. Machine learning for prediction of sustained opioid prescription after anterior cervical discectomy and fusion. *Spine J* . 2019;19(6):976-983. doi:10.1016/j.spinee.2019.01.009
108	Reid DBC, Shah KN, Shapiro BH, Ruddell JH, Akelman E, Daniels AH. Mandatory Prescription Limits and Opioid Utilization Following Orthopaedic Surgery. *J Bone Jt Surg - Am Vol* . 2019;101(10):1-8. doi:10.2106/JBJS.18.00943
109	Clarke H, Soneji N, Ko DT, Yun L, Wijeysundera DN. Rates and risk factors for prolonged opioid use after major surgery: Population based cohort study. *BMJ* . 2014;348(February):1-10. doi:10.1136/bmj.g1251
110	Berger I, Strother M, Talwar R, et al. National variation in opioid prescription fills and long-term use in opioid naive patients after urological surgery. *J Urol* . 2019;202(5):1036-1042. doi:10.1097/JU.0000000000000343
111	Orfield NJ, Gaddis A, Russell KB, Hartman DW, Apel PJ, Mierisch C. New Long-Term Opioid Prescription-Filling Behavior Arising in the 15 Months After Orthopaedic Surgery. *J Bone Joint Surg Am* . 2020;102(4):332-339. doi:10.2106/JBJS.19.00241
112	Clement KC, Canner JK, Whitman GJR, Lawton JS, Grant MC, Sussman MS. New Persistent Opioid Use After Aortic and Mitral Valve Surgery in Commercially Insured Patients. *Ann Thorac Surg* . 2020;110(3):829-835. doi:10.1016/j.athoracsur.2019.12.031
113	Brummett CM, Waljee JF, Goesling J, et al. New persistent opioid use after minor and major surgical procedures in US adults. *JAMA Surg* . 2017;152(6). doi:10.1001/jamasurg.2017.0504
114	Tam CA, Dauw CA, Ghani KR, et al. New Persistent Opioid Use After Outpatient Ureteroscopy for Upper Tract Stone Treatment. *Urology* . 2019;134:103-108. doi:10.1016/j.urology.2019.08.042
115	Santosa KB, Hu HM, Brummett CM, et al. New persistent opioid use among older patients following surgery: A Medicare claims analysis. *Surg (United States)* . 2020;167(4):732-742. doi:10.1016/j.surg.2019.04.016
116	Xu Y, Cuthbert CA, Karim S, et al. Associations Between Physician Prescribing Behavior and Persistent Postoperative Opioid Use Among Cancer Patients Undergoing Curative-intent Surgery. *Ann Surg* . 2020;Publish Ah(Xx):1-6. doi:10.1097/sla.0000000000003938
117	Lee JSJ, Hu HM, Edelman AL, et al. New persistent opioid use Among patients with cancer after curative-intent surgery. *J Clin Oncol* . 2017;35(36):4042-4049. doi:10.1200/JCO.2017.74.1363
118	Finney FT, Gossett TD, Hu HM, et al. New Persistent Opioid Use Following Common Forefoot Procedures for the Treatment of Hallux Valgus. *J Bone Jt Surg - Am Vol* . 2019;101(8):722-729. doi:10.2106/JBJS.18.00793
119	Kang C, Shu X, Herrell SD, Miller NL, Hsi RS. Opiate Exposure and Predictors of Increased Opiate Use after Ureteroscopy. *J Endourol* . 2019;33(6):480-485. doi:10.1089/end.2018.0796
120	Inacio MCS, Hansen C, Pratt NL, Graves SE, Roughead EE. Risk factors for persistent and new chronic opioid use in patients undergoing total hip arthroplasty: A retrospective cohort study. *BMJ Open* . 2016;6(4):1-10. doi:10.1136/bmjopen-2015-010664
121	Sharma M, Ugiliweneza B, Sirdeshpande P, Wang D, Boakye M. Opioid Dependence and Health Care Utilization after Decompression and Fusion in Patients with Adult Degenerative Scoliosis. *Spine (Phila Pa 1976)* . 2019;44(4):280-290. doi:10.1097/BRS.0000000000002794
122	Ren M, Bryant BR, Harris AB, et al. Opioid use after adult spinal deformity surgery: Patterns of cessation and associations with preoperative use. *J Neurosurg Spine* . 2020;33(4):490-495. doi:10.3171/2020.3.SPINE20111
123	Chen MJ, Zhang S, Debaun M, Fogel N, Bishop JA, Gardner MJ. Opioid use after ankle fracture surgery: Current trends in the United States. *Curr Orthop Pract* . 2019;30(4):332-335. doi:10.1097/BCO.0000000000000764
124	Tran T, Castello J, Taylor SE, et al. Opioid Use and Appropriateness of Supply After Total Knee or Hip Arthroplasty: An Australian Perspective. *J Am Acad Orthop Surg* . 2020;28(23):e980-e989. doi:10.5435/jaaos-d-19-00789
125	Pugely AJ, Bedard NA, Kalakoti P, et al. Opioid use following cervical spine surgery: trends and factors associated with long-term use. *Spine J* . 2018;18(11):1974-1981. doi:10.1016/j.spinee.2018.03.018
126	Khazi ZM, Lu Y, Shamrock AG, Duchman KR, Westermann RW, Wolf BR. Opioid use following shoulder stabilization surgery: risk factors for prolonged use. *J Shoulder Elb Surg* . 2019;28(10):1928-1935. doi:10.1016/j.jse.2019.05.026
127	Kalakoti P, Hendrickson NR, Bedard NA, Pugely AJ. Opioid Utilization Following Lumbar Arthrodesis Trends and Factors Associated With Long-term Use. *Spine (Phila Pa 1976)* . 2018;43(17):1208-1216. doi:10.1097/BRS.0000000000002734
128	Sabesan VJ, Chatha K, Goss L, Ghisa C, Gilot G. Can patient and fracture factors predict opioid dependence following upper extremity fractures?: A retrospective review. *J Orthop Surg Res* . 2019;14(1):1-5. doi:10.1186/s13018-019-1233-7
129	Nguyen A V., Ross E, Westra J, et al. Opioid Utilization in Geriatric Patients after Operation for Degenerative Spine Disease. *J Neurosurg Anesthesiol* . 2020;(Online ahead of print). doi:10.1097/ANA.0000000000000682
130	Reid DBC, Patel SA, Shah KN, et al. Opioid-limiting legislation associated with decreased 30-day opioid utilization following anterior cervical decompression and fusion. *Spine J* . 2020;20(1):69-77. doi:10.1016/j.spinee.2019.08.014